



UNIVERSITY OF THE PUNJAB

AD (2 Years) & BS (4 Years) Degree Program / First Semester – Fall 2024

Paper: Calculus

Course Code: SMTH-101

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Give short answer to the following concepts: (5x6=30)

1. If $y = x^3 + 2x + 3$, the find $\frac{dy}{dx}$.
2. Examine the function defined as $f(x) = x^3 - 6x^2 + 9x$ for extreme values.
3. Show that $y = x^e$ has minimum value at $x = \frac{1}{e}$, if possible; otherwise, provide a justification..
4. Show that $\varphi(x) = x$ is continuous on $\mathbb{R}$.
5. Solve $\int \ln x dx$, if possible; otherwise, provide a justification.

Q.2. Answer the following questions. (3x10=30)

1. Solve $\int \sin^4 x dx$, if possible; otherwise, provide a justification.
2. Sketch the graph of the function $y = \sin 2x$.
3. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, then find $f \circ g$ and its domain.